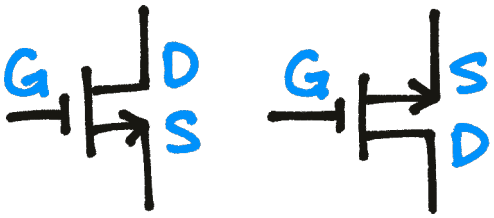
NMOS

Schematic



Equations

Equation	Conditions	Notes
$i_D = 0$	$v_{GS} < V_{tn}$	(Cutoff region)
$i_D = k'_n \left(\frac{W}{L}\right) \left(v_{OV} - \frac{1}{2} v_{DS}\right) v_{DS}$	$v_{DS} < v_{OV}$	(Triode region)
$i_D = \frac{1}{2} k'_n \left(\frac{W}{L}\right) v_{OV}^2$	$v_{DS} \geq v_{OV}$	(Saturation region)

Notes